Juan D. Beltran, DPhil

+447922857788 id.beltran43@gmail.com
London, UK JuanDBeltranP
bit.ly/jdbeltrancv jdbeltran

General Skills

Artificial Intelligence | Deep Learning | Strategic Thinking | Machine Learning | Quantitative Research | Commodities | Leadership | Business Intelligence | Geospatial Analysis | Cloud Computing | Time Series Analysis | Automation | Data Visualisation | Communication | Project Management | Negotiation | Team player | Agile Methodologies | Machine Learning Operations - MLOps

Technical Skills

Advanced Programming: Python, SQL, R
Specialized Libraries: Pandas, GeoPandas, Scikit-Learn, TensorFlow, PyTorch, Prophet
Machine Learning: Deep Learning, Time series analysis & forecasting (model selection, feature engineering, deployment)
GIS & Geospatial Analysis: GIS modeling, geospatial machine learning (ArcGIS, QGIS, Python libraries)
Data Automation & BI: Power BI, Tableau, Bash, CRON, custom Python/R scripts
Leadership & Management: Experience leading data science teams and projects, and deploying end-to-end machine learning solutions to production
Big Data & Cloud: BigQuery, AWS, and Spark

Languages

- English (Fluent)
- Spanish (Native)

Interests

- Sustainability and Carbon Markets
- Gym and Training
- Travelling

Awards

Louis Dreyfus-Weidenfeld Scholarship and Leadership Programme at University of Oxford — Weidenfeld-Hoffmann Trust

Lead Data Scientist

Summary

Results-oriented Lead Data Scientist with 7+ years of experience applying advanced analytics and geospatial modeling to solve complex challenges. Proven expertise in leveraging machine learning, time-series forecasting, and GIS analysis, with a focus on commodities trading and environmental impact assessment. Demonstrated success in developing and deploying scalable ML solutions using Python, PyTorch, and cloud platforms. Passionate about combining technical skills with a deep commitment to nature conservation, seeking opportunities to drive meaningful change through data-driven innovation.

Experience

Data Scientist Consultant | Freelancer | May 2024 – Aug 2024

- Provided consulting expertise to commodities research teams, developing and implementing machine learning models to automate data analysis and enhance predictive capabilities.

Lead Data Scientist | ECOM Agroindustrial Corp. Ltd. | Dec 2022 – May 2024

- Pioneered a global deforestation risk model leveraging satellite imagery, machine learning, and GIS. Directly contributed to ECOM's carbon footprint assessment.
- Led the data science team to deliver impactful end-to-end data products & ML models for cocoa, coffee, & cotton market insights.
- Engineered algorithmic trading signals models for cocoa, coffee, & cotton, integrating fundamental & technical analysis.
- Developed, Deployed and Maintained the predictive price forecasting models for coffee, cotton, & cocoa, enhancing decision-making.
- Championed data-driven culture across the organization, promoting informed decision-making at all levels.

Senior Data Scientist | ECOM Agroindustrial Corp. Ltd. | Dec 2019 – Dec 2022

- Implemented an automated alert system to identify anomalies in the futures markets for cocoa, coffee and cotton.
- Built a weather platform to assess temperature and precipitation anomalies in the regions of interest for cocoa, coffee and cotton markets.
- Assessed future climate change impacts on global coffee and cocoa plantations around the world.
- Leveraged satellite imagery and machine learning for accurate crop yield predictions.
- Led development of models to predict the outcome of key market reports for cotton yields and production estimations.

Product Developer & Data Science | Decoded | Oct 2018 – Dec 2019

- Co-developed 40%+ of the curriculum for the *Data Academy*, a program designed to integrate data science into blue chip companies' operations.
- Collaborated with key stakeholders at leading financial institutions and retailers, including Societe Generale, UBS, Unilever, Nike, and M&S.
- Developed comprehensive educational materials on data science concepts, tools, and techniques (Python/R, statistics, ML, time series, text analysis).
- Successfully empowered employees at various corporations through data science training.

Postdoctoral Researcher | University College London | Jan 2018 – Oct 2018

- Developed a real time model to track the Zika Virus spread in Brazil.

Education

- PhD (DPhil) in Plant Sciences - University of Oxford (2013-2017)
- MSc in Biology (1st place) - Universidad de los Andes, Colombia (2012)
- BSc in Biology - Universidad de los Andes, Colombia (2010)